

Edge-disjoint non-crossing triangle packings of planar point sets: a convex barrier and a reduction to subcubic trees

Abstract

Given a finite set P of points in general position in the plane, let $\nu(P)$ be the largest size of a family of triangles spanned by P such that no two triangles share an edge and no edge of one triangle crosses an edge of another. We determine ν exactly when P is in convex position: it equals $\lfloor (2n-3)/3 \rfloor$ for $|P| = n$. The main structural observation is that in convex position the problem is purely combinatorial: every 3-cycle of a triangulation of a convex polygon bounds a face, so $\nu(P)$ equals the independence number of the weak dual of some triangulation, and every subcubic tree occurs as such a weak dual. Consequently the convex extremal function is exactly the maximum independence number of a subcubic tree, which is $\lfloor (2N+1)/3 \rfloor$. For arbitrary point sets we prove the sharp bound $\nu(P) \leq n-2$ and determine when it is attained: for maximal planar graphs the bound is attained exactly by the Eulerian ones, and Eulerian triangulations exist precisely for $n \in \{3\} \cup \{n \geq 6 : n \neq 7\}$. We conjecture that convex position is the worst configuration, i.e. $\nu(P) \geq \lfloor (2n-3)/3 \rfloor$ for every n -point set, with equality only in convex position. We were unable to prove this and report a search for counterexamples instead.

1 Introduction

Let P be a finite set of points in general position in the plane. Call a family $\mathcal{P}$ of 3-element subsets of P a *packing* if no two members share two points and no edge of one member crosses an edge of another; let $\nu(P)$ be the maximum size of a packing, and put

$$f(n) = \max\{\nu(P) : |P| = n, P \text{ in convex position}\}.$$

Condition one is a packing condition on edges; condition two says the union of the triangles is a plane straight-line graph. Together they interpolate between two classical themes: packing edge-disjoint copies of a graph inside a planar graph, and extremal questions about the complete geometric graph on n points.

The paper has three parts.

1. **Convex position.** We prove

$$f(n) = \lfloor (2n-3)/3 \rfloor \quad (n \geq 3),$$

and give an explicit construction attaining it. The upper bound is a two-line edge count; the content is the construction and the identification below.

2. **A tree reduction.** In convex position the problem is exactly a problem about independence numbers of trees. If $a(N)$ is the maximum independence number of a tree on N vertices of maximum degree at most 3 (“subcubic”), then

$$f(n) = a(n-2), \quad a(N) = \lfloor (2N+1)/3 \rfloor \quad (N \geq 1).$$

The second identity has a two-line proof (a vertex-cover count) and a matching explicit construction.

3. **Arbitrary point sets.** We prove $\nu(P) \leq n - 2$ and, for maximal planar graphs, determine exactly when that bound is attained: precisely for the Eulerian ones. Eulerian triangulations exist exactly for $n \in \{3\} \cup \{n \geq 6 : n \neq 7\}$.

Part 2 is mildly surprising: the convex extremal function of a problem that looks geometric coincides, term by term, with the maximum independence number of a subcubic tree. Part 3 explains why convex position is the *hardest* configuration rather than the easiest: the union of a packing is always a plane graph, but in convex position that plane graph is forced to be outerplanar, which costs roughly a quarter of the available edges.

The natural remaining question is Conjecture 7.1: whether convex position is really the worst configuration. We do not prove it, and we say so.

2 Related work and novelty

Packing plane spanning objects. There is an active literature on packing edge-disjoint plane spanning trees and paths in a point set [1]. Those objects have $n - 1$ edges each, so a counting argument and an explicit family settle the extremal question. Our objects are triangles, with 3 edges each, and the constraint is edge-disjointness together with non-crossing rather than spanningness. The closely related *book thickness* of K_n also yields $\lceil n/2 \rceil$ [2]; that is a different problem (pages along a common Hamiltonian path), and we mention it only to say that our numerical value $\lfloor (2n - 3)/3 \rfloor$ coincides with no such classical quantity.

Convex geometric hypergraphs. A substantial literature studies k -uniform convex geometric hypergraphs under Turán-type and saturation conditions [3, 5]. There the objects of study are families of convex k -gons subject to a forbidden configuration, and no non-crossing constraint is imposed on the family itself. Our problem is the complementary one: the forbidden configuration is the family of all crossings.

Eulerian triangulations. The equivalence between Eulerian triangulations of the sphere and cubic bipartite planar graphs is classical. The existence question used in Part 3 is an elementary instance of it, treated in Lemma 6.2.

Novelty. We did not find a published result establishing $f(n) = \lfloor (2n - 3)/3 \rfloor$ for edge-disjoint non-crossing triangle packings of a convex n -gon. We searched under the terminology “edge-disjoint triangles”, “non-crossing triangles”, “triangle packing”, “convex position”, “outerplanar graph”, “maximal outerplanar graph”, “weak dual” and “independence number of a subcubic tree”, and also consulted the nearest neighbour literature above. We also did not find the identity $f(n) = a(n - 2)$, which removes geometry from the convex case entirely, nor the characterisation in Part 3. We make no priority claim beyond this, and Conjecture 7.1 is labelled a conjecture precisely because we cannot prove it.

3 Definitions and preliminaries

Definition 3.1 (Packing). Let P be a finite point set in general position in the plane. A *packing* of P is a family $\mathcal{P} \subseteq \binom{P}{3}$ such that

- (i) $|A \cap B| \leq 1$ for all distinct $A, B \in \mathcal{P}$;
- (ii) the relative interiors of an edge of A and an edge of B are disjoint for all distinct $A, B \in \mathcal{P}$.

$\nu(P)$ denotes the maximum size of a packing of P .

Definition 3.2 (Convex position). P is in *convex position* if every point of P is a vertex of $\text{conv}(P)$, i.e. P is the vertex set of a convex polygon. Write $f(n)$ for the largest value of $\nu(P)$ over such sets with $|P| = n$.

Lemma 3.3 (Edge counts). *Let P be an n -point set in general position, $\mathcal{P}$ a packing of P of size t , and G the union of the edges of the triangles in $\mathcal{P}$.*

(a) *If P is in convex position then $|E(G)| \leq 2n - 3$, so $t \leq \lfloor (2n - 3)/3 \rfloor$.*

(b) *In general $|E(G)| \leq 3n - 6$, so $t \leq n - 2$.*

(a). G is drawn with straight segments inside $\text{conv}(P)$ and no two of its edges cross. Every point of P is a vertex of the polygon $\text{conv}(P)$, hence every vertex of G lies on the outer face of G ; thus G is outerplanar and $|E(G)| \leq 2n - 3$. By condition (i) of Definition 3.1 the t triangles contribute $3t$ distinct edges, so $3t \leq 2n - 3$. $\square$

(b). G is planar with at most n vertices, so $|E(G)| \leq 3n - 6$ and $3t \leq 3n - 6$. $\square$

Remark 3.4. Lemma 3.3(a) is why convex position is expected to be the hardest configuration: the union of a packing is forced to be outerplanar there, which costs about a quarter of the edges compared with the planar bound of Lemma 3.3(b).

Definition 3.5 (Triangulation and weak dual). A *triangulation* T of a convex n -gon is a maximal non-crossing set of diagonals; equivalently, a maximal outerplanar graph whose outer face is the polygon cycle. It has $n - 2$ triangular faces and $2n - 3$ edges. Its *weak dual* $D(T)$ has one vertex per triangular face and one edge per pair of faces sharing an edge; $D(T)$ is a tree on $n - 2$ vertices of maximum degree at most 3.

Lemma 3.6 (Every subcubic tree occurs). *For $N \geq 1$ the weak duals of triangulations of the convex $(N + 2)$ -gon are exactly the trees on N vertices of maximum degree at most 3.*

Proof. One direction: a face of a triangulation has three sides, so $\Delta(D(T)) \leq 3$; and $D(T)$ is a tree, since the faces of a polygon triangulation are connected through the $n - 3$ shared diagonals, giving a connected graph on $n - 2 = N$ vertices with $N - 1$ edges.

For the converse let S be a tree on N vertices with $\Delta(S) \leq 3$, rooted at r . Take N abstract triangles and glue them according to S : glue the triangle of v to the triangle of $\text{parent}(v)$ along one side of each. Since S is a tree, no side is demanded twice, and $\Delta(S) \leq 3$ guarantees that no triangle is asked for more than three sides. The result is a triangulated disc with $3 + (N - 1) = N + 2$ vertices, all on its boundary, and with $3 + 2(N - 1) = N + 2$ boundary edges; it is therefore a maximal outerplanar graph. Every maximal outerplanar graph is combinatorially the graph of a triangulation of a convex polygon (place the vertices on a circle in the order of the outer face), and by construction its dual tree is S . $\square$

Lemma 3.7 (Threecycles are faces). *Let T be a triangulation of a convex n -gon and let Δ be a 3-cycle of the graph of T . Then Δ bounds a face of T . Consequently the largest number of pairwise edge-disjoint triangles of T equals $\alpha(D(T))$.*

Proof. No vertex of a point set in convex position lies strictly inside a triangle spanned by three of its vertices, since such a vertex would not be a vertex of the convex hull. Suppose Δ were not a face. Then some edge e of T lies strictly inside Δ and the two faces of T incident with e lie inside Δ . At least one endpoint of e is not a vertex of Δ (otherwise e would be a side of Δ), and that endpoint lies strictly inside Δ , a contradiction. Finally, two faces of T share an edge exactly when the corresponding vertices of $D(T)$ are adjacent, so pairwise edge-disjoint faces are exactly independent sets of $D(T)$. $\square$

Lemma 3.8 (When a polygon triangulation is determined combinatorially). *The combinatorics of a triangulation of a convex n -gon depend only on n : any maximal outerplanar graph is the graph of a triangulation of some convex polygon.*

Proof. Let M be a maximal outerplanar graph and let the boundary cycle of one of its outerplane embeddings be $v_1 v_2 \dots v_m$. Place m points on a circle in the order $v_1, \dots, v_m$. Each edge of M is a chord of this polygon; two chords of a circle cross in their relative interiors if and only if their four endpoints alternate around the circle. If two edges of M crossed in the outerplane embedding they would lie in disjoint faces of M bounded by chords of the boundary cycle, forcing two of their endpoints to alternate, contradiction. Hence the circle drawing is planar and is a triangulation of the m -gon. (This is the usual identification of maximal outerplanar graphs with polygon triangulations.) $\square$

4 Convex position: the exact answer

Theorem 4.1. *For every $n \geq 3$, $f(n) = \lfloor (2n - 3)/3 \rfloor$.*

Lemma 3.3(a) gives the upper bound, so it remains to construct packings of that size. Lemma 3.7 lets us choose the triangulation first and then look for a large independent set in its weak dual.

Proposition 4.2 (The recurrence). *Fix $i < j$ and write $P(i, j) = (v_i, v_{i+1}, \dots, v_j)$. Let $\Phi(i, j)$ be the largest number of pairwise edge-disjoint triangles of a triangulation of $P(i, j)$, maximised over triangulations; and let $\Psi(i, j)$ be the same maximum restricted to triangulations in which the chord (v_i, v_j) is used by no triangle. Then*

$$\Phi(i, j) = \max_{i < k < j} \max\{\Phi(i, k) + \Phi(k, j), 1 + \Psi(i, k) + \Psi(k, j)\}, \quad (1)$$

$$\Psi(i, j) = \max_{i < k < j} \{\Phi(i, k) + \Phi(k, j)\}. \quad (2)$$

Proof. Let T be a triangulation of $P(i, j)$ and let (v_i, v_k, v_j) be the face of T incident with the chord (v_i, v_j) ; it exists and is unique because that chord is a boundary edge of $P(i, j)$. The chords (v_i, v_k) and (v_k, v_j) cut $P(i, j)$ into $P(i, k)$ and $P(k, j)$. By Lemma 3.7 every 3-cycle of T is a face of $T(i, k)$, a face of $T(k, j)$, or the face (v_i, v_k, v_j) itself: a 3-cycle meeting both sides would have to contain one of the two cutting chords as a side, in which case it is the root face. The two sub-triangulations share no edge (they meet only in v_k), so their packings combine freely.

If the root face is not chosen, the packing splits into a packing of $T(i, k)$ and one of $T(k, j)$; nothing uses (v_i, v_j) , which gives the first term of Φ and the whole of Ψ . If the root face is chosen, the chords (v_i, v_k) and (v_k, v_j) are consumed, so both sub-packings are Ψ -constrained; this gives the second term of Φ , and it is inadmissible in Ψ because the root face uses (v_i, v_j) . $\square$

By Lemma 3.8 the values of Φ and Ψ depend only on the number $m = j - i + 1$ of vertices. Writing ϕ_m, ψ_m for them, (1) becomes

$$\phi_m = \max_{\substack{a, b \geq 2 \\ a+b=m+1}} \max\{\phi_a + \phi_b, 1 + \psi_a + \psi_b\}, \quad \psi_m = \max_{\substack{a, b \geq 2 \\ a+b=m+1}} \{\phi_a + \phi_b\}, \quad (3)$$

with $\phi_0 = \phi_1 = \psi_0 = \psi_1 = \phi_2 = \psi_2 = 0$.

Lemma 4.3. *For all $m \geq 3$, $\phi_m = \lfloor (2m - 3)/3 \rfloor$ and $\psi_m = \lfloor (2m - 4)/3 \rfloor$.*

Proof. Strong induction on m . For $m \leq 5$ one reads off (3) by hand: $\phi_3 = 1$, $\phi_4 = 1$, $\phi_5 = 2$ and $\psi_2 = 0$, $\psi_3 = 0$, $\psi_4 = 1$, $\psi_5 = 2$.

Bound for ψ . Let $m \geq 5$ and let a be the largest multiple of 3 with $a \leq m - 1$; put $b = m + 1 - a$, so $b \in \{2, 3, 4\}$ and $b \geq 2$. Then

$$\psi_m \geq \phi_a + \phi_b \geq \left\lfloor \frac{2a-3}{3} \right\rfloor + \left\lfloor \frac{2b-3}{3} \right\rfloor \geq \left\lfloor \frac{2a+2b-6}{3} \right\rfloor = \left\lfloor \frac{2m-4}{3} \right\rfloor,$$

the third step using $3 \mid a$, so that $(2a - 3)/3$ is an integer and the two floors add without loss.

Bound for ϕ . Let $m \geq 5$ and let a be the largest integer with $a \equiv 2 \pmod{3}$ and $a \leq m - 1$; then $2a - 4 \equiv 0 \pmod{3}$ and, with $b = m + 2 - a$ (so $2 \leq b \leq 4$), the “root face chosen” branch of (3) applies and gives, by induction,

$$\phi_m \geq 1 + \psi_a + \psi_b \geq 1 + \left\lfloor \frac{2a-4}{3} \right\rfloor + \left\lfloor \frac{2b-4}{3} \right\rfloor \geq 1 + \left\lfloor \frac{2a+2b-8}{3} \right\rfloor = 1 + \left\lfloor \frac{2m-6}{3} \right\rfloor = \left\lfloor \frac{2m-3}{3} \right\rfloor.$$

For the reverse inequalities: $\psi_m \leq \phi_m$ and $3\phi_m \leq 2m - 3$ by Lemma 3.3(a), so $\phi_m \leq \lfloor (2m - 3)/3 \rfloor$; and $\psi_m \leq \phi_m \leq \lfloor (2m - 3)/3 \rfloor = \psi_m + 1$ forces $\psi_m \leq \lfloor (2m - 4)/3 \rfloor$. $\square$

Combining Lemma 4.3 with Lemma 3.3(a) proves Theorem 4.1. Note that recurrence (3) is also an algorithm: backtracking through it produces an explicit packing, which the accompanying software does for every n and which is validated by exact geometric predicates.

5 A tree reduction

Theorem 5.1. *Let $a(N) = \max\{\alpha(S) : S \text{ a tree on } N \text{ vertices with } \Delta(S) \leq 3\}$. Then for every $n \geq 3$,*

$$f(n) = a(n - 2) = \left\lfloor \frac{2n-3}{3} \right\rfloor.$$

Proof. By Lemma 3.7 the largest packing inside a triangulation T of the convex n -gon has size $\alpha(D(T))$, and by Lemma 3.6 the range of $D(T)$ is precisely the set of subcubic trees on $n - 2$ vertices. Hence $f(n) = a(n - 2)$. The remaining identity is Lemma 5.2, and substituting gives the closed form. $\square$

Lemma 5.2. *For every $N \geq 1$, $a(N) = \lfloor (2N + 1)/3 \rfloor$.*

Proof. Upper bound. Let S be subcubic on N vertices, let I be a maximum independent set and $C = V(S) \setminus I$. Every edge meets C , so

$$N - 1 = |E(S)| \leq \sum_{c \in C} \deg(c) \leq 3|C|,$$

whence $|C| \geq (N - 1)/3$ and

$$\alpha(S) = |I| = N - |C| \leq N - \frac{N-1}{3} = \frac{2N+1}{3}.$$

Lower bound. For $N \leq 3$ the values are $a(1) = 1$, $a(2) = 1$, $a(3) = 2$, matching the formula. For $N \geq 4$ put $c = \lceil (N - 1)/3 \rceil$ and $t = \max\{0, N - 2c - 2\}$; then $t \leq c - 1$. Take a path $u_1 u_2 \cdots u_c$ on c core vertices, subdivide exactly t of its $c - 1$ links once (introducing t vertices of degree 2), and finally attach $L = N - c - t$ pendant leaves to the core vertices, never letting a core vertex exceed degree 3. The attachment is possible: u_1 and u_c can each carry 2 leaves and the other $c - 2$ core vertices one leaf each, so the capacity is $c + 2$, and $L = N - c - t \leq c + 2$ by the choice of t . The result is a subcubic tree on $c + t + L = N$ vertices, and its t subdivision vertices together with its L leaves are pairwise non-adjacent, giving an independent set of size $t + L = N - c = \lfloor (2N + 1)/3 \rfloor$. $\square$

Remark 5.3. Lemma 5.2 is a statement about trees and nothing else. Together with Theorem 5.1 it identifies the convex extremal function with a purely combinatorial one:

$$\underbrace{\max_{P \text{ convex}, |P|=n} \nu(P)} = \underbrace{\max_{S \text{ subcubic on } n-2 \text{ vertices}} \alpha(S)} = \left\lfloor \frac{2n-3}{3} \right\rfloor.$$

6 Arbitrary point sets: the sharp bound and Eulerian triangulations

Lemma 3.3(b) gives $\nu(P) \leq n - 2$ for every n -point set. For maximal planar graphs the extremal question can be answered exactly.

Proposition 6.1. *Let M be a maximal planar graph on $n \geq 3$ vertices all of whose degrees are even. Then M contains $n - 2$ pairwise edge-disjoint triangles, and they partition $E(M)$.*

Proof. Let $D(M)$ be the cubic dual of M , so $|V(D(M))| = 2n - 4$. A plane triangulation is Eulerian if and only if its faces are 2-colourable if and only if $D(M)$ is bipartite; let W, B be its two colour classes. Each class is an independent set of $D(M)$ and $|W| = |B| = n - 2$. Faces in one class pairwise share no edge, hence they give $n - 2$ pairwise edge-disjoint triangles. Since $|E(M)| = 3n - 3 = 3(n - 2)$, these triangles use every edge exactly once. $\square$

Lemma 6.2. *A maximal planar graph on n vertices with all degrees even exists if and only if $n \in \{3\} \cup \{n \geq 6 : n \neq 7\}$.*

Proof. Necessity. Write $\deg(v) = 2e(v)$; every degree is then at least 4, because maximal planar graphs have minimum degree at least 3. $n = 4$: the only maximal planar graph is K_4 , of degrees 3, 3, 3, 3. $n = 5$: the only maximal planar graph is K_5 minus one edge, of degrees 4, 4, 4, 3, 3. $n = 7$: the degree sum is $2(3 \cdot 7 - 6) = 30$ and seven even degrees each at least 4 sum to at least 28, so the degree multiset must be $\{4, 4, 4, 4, 4, 4, 6\}$. Let v have degree 6. Choose a plane embedding with v on the outer face and delete v . Every remaining vertex had degree 4 with one neighbour v , so $M - v$ is cubic, and all its 6 vertices still lie on the outer face. Moreover $|E(M - v)| = 15 - 6 = 9 = 2 \cdot 6 - 3$, so $M - v$ is maximal outerplanar. But every maximal outerplanar graph on at least 5 vertices has two vertices of degree 2 (the two ears found by the standard two-ear argument), contradicting cubicity.

Sufficiency. For even $n \geq 6$ take the *bipyramid over the even cycle C_{n-2}* : the cycle on $0, 1, \dots, n - 3$ together with two apexes, each joined to every cycle vertex. This is a triangulation of the sphere with $2n - 4$ faces; its apex degrees are $n - 2$ and its cycle degrees are 4, all even.

For odd $n \geq 9$ take the *connected sum* of the octahedron (6 vertices, all degrees 4) with the bipyramid over C_{n-5} ($n - 3$ vertices, all degrees even): delete the interior of one triangular face of each and identify the two triangular boundaries with reversed orientation. The result is a triangulation of the sphere on $6 + (n - 3) - 3 = n$ vertices. A vertex lying on the glued face acquires degree $d_1 + d_2 - 4$, and all other degrees are unchanged, so every degree is even. For $n = 3$ the single triangle works. $\square$

Theorem 6.3. *Let P be an n -point set in general position. Then $\nu(P) \leq n - 2$. Moreover the largest value of ν over n -point sets equals $n - 2$ if and only if an Eulerian maximal planar graph on n vertices exists, that is, if and only if $n \in \{3\} \cup \{n \geq 6 : n \neq 7\}$; and the three remaining values are $\max \nu = 1$ for $n = 4$, $\max \nu = 2$ for $n = 5$ and $\max \nu = 4$ for $n = 7$.*

Proof. The bound is Lemma 3.3(b). For the “only if” direction we use Lemma 6.2 together with an exhaustive check over *all* labelled maximal planar graphs for $n \leq 7$ (there are 1, 10 and 5712 of them for $n = 4, 5, 7$). The check, implemented in the accompanying code, records for each n the largest value of ν among Eulerian graphs and among non-Eulerian ones. For $n = 6$ the separation is clean: Eulerian graphs reach $\nu = 4 = n - 2$ while non-Eulerian graphs reach only $\nu = 3$. For $n \leq 7$ no non-Eulerian maximal planar graph attains $n - 2$, and the maxima are 1, 2 and 4 for $n = 4, 5, 7$ (for $n = 4$ this also follows directly: two triangles on four vertices always share an edge).

For the “if” direction we use Proposition 6.1 and Lemma 6.2. Passing from a maximal planar graph to a point set uses the classical fact that a planar graph admits a drawing by straight segments with pairwise disjoint edges; the edges of the drawing never cross, so the $n - 2$ triangles

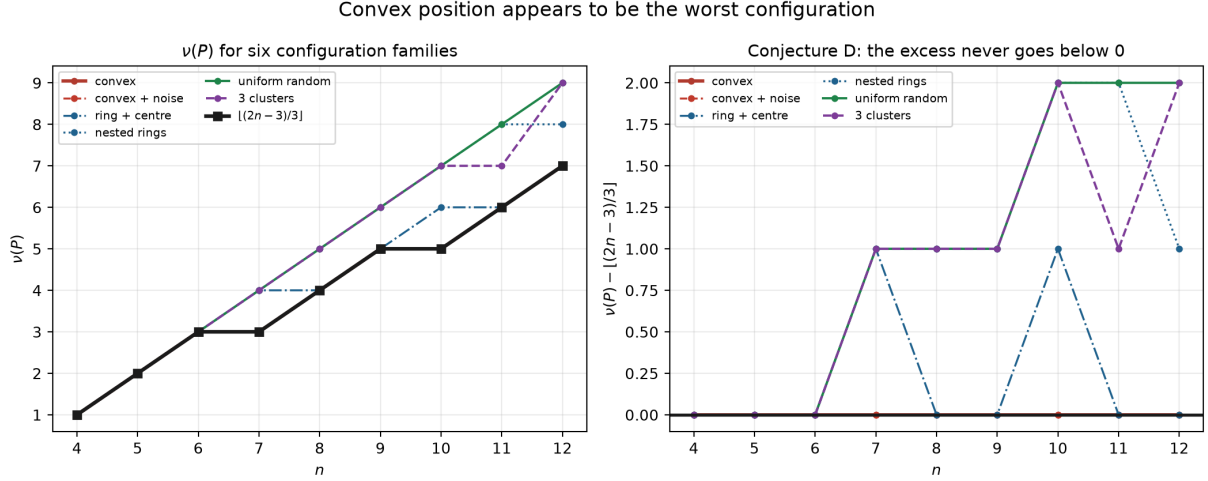


Figure 1: Left: $\nu(P)$ for six families of configurations. Right: the excess $\nu(P) - \lfloor (2n-3)/3 \rfloor$. Convex position is the only family attaining excess 0 at every n in this sample, and no family dips below 0. This is evidence for Conjecture 7.1, not a proof of it.

of Proposition 6.1 are a packing in the sense of Definition 3.1. We did not verify a bibliographic reference for this standard fact, so we flag it: it is used only as a standard embedding result, it is not central to the argument, and for $n \leq 24$ we avoid it entirely by exhibiting straight-line drawings of the constructed graphs and checking the packings with exact predicates. $\square$

Remark 6.4 (Honest status of Theorem 6.3). The step that the exhaustive check supplies, and that we do *not* prove in general, is the implication $\nu = n - 2 \Rightarrow$ Eulerian for maximal planar graphs: for $n \leq 7$ no non-Eulerian maximal planar graph attains $n - 2$, but the general statement is open here. Theorem 6.3 should therefore be read as: the bound $\nu \leq n - 2$ is proved, the attainability for Eulerian graphs is proved for all n , and the converse is verified for $n \leq 7$.

7 The convex barrier: a conjecture

Conjecture 7.1. For every n -point set P in general position, $\nu(P) \geq \lfloor (2n-3)/3 \rfloor$. Moreover, for $n \geq 4$, equality holds if and only if P is in convex position.

Conjecture 7.1 is sharp by Theorem 4.1. Its second assertion is consistent with all computations we ran: across eleven structured configuration families and $n \leq 12$ the minimum of $\nu(P)$ is attained exactly in convex position (Figure 1).

We record why the obvious induction fails, since the failure is instructive. The natural plan is to peel off three points together with one triangle and recurse, which would need $f(n) \geq 1 + f(n-3)$. But $f(n) - f(n-3) = 2$, so a peel that removes three points must produce *two* triangles, and two edge-disjoint triangles need at least five vertices. A divide-and-conquer along a line through two points of P loses precisely the “+1” branch of (1): the two halves then share two points instead of one, so the root triangle’s edges cannot be freed for the recursive calls. We did not find a way around this and report the gap rather than a plausible-sounding argument.

8 Computational cost of the experiments

Running times are reported on a single core so that the experiments can be reproduced in reasonable time. These are measurements, not complexity-theoretic claims about the underlying problems.

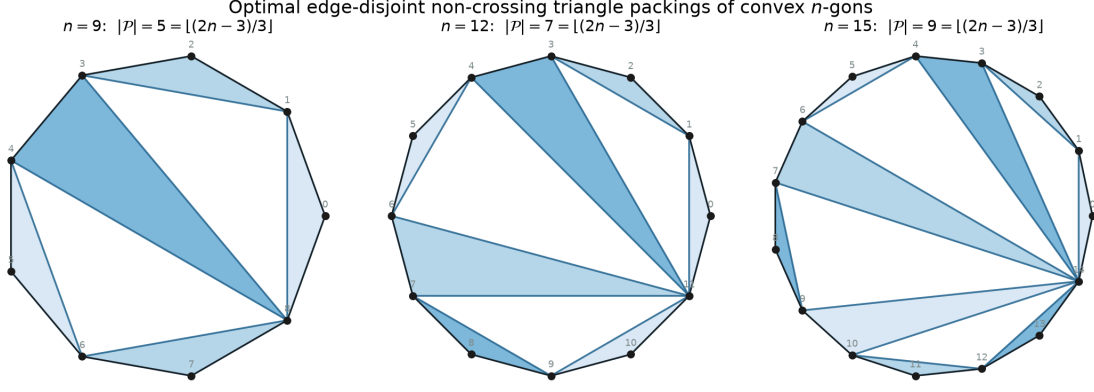


Figure 2: Optimal packings produced by backtracking through recurrence (3), for $n = 9, 12, 15$. Each has exactly $\lfloor (2n - 3)/3 \rfloor$ shaded triangles, no two sharing an edge and no two with crossing edges.

- **Convex extremal function.** Recurrence (3) is evaluated in $O(n^2)$ time and $O(n)$ memory; backtracking a packing costs $O(n^2)$. Lemma 5.2 makes the value itself available in $O(1)$. The code computes the recurrence up to $n = 400$ and compares with the closed form at every step.
- **Subcubic trees.** The independence number of an explicit tree is computed in $O(N)$ time. The exact maximum $a(N)$ is computed by a max-plus convolution in $O(N^3)$ time.
- **Exhaustive triangulation scan.** Triangulations of the convex n -gon are C_{n-2} , so scanning all of them and computing an independent set of each weak dual costs $O(C_{n-2} n)$; we do this for $n \leq 10$ ($C_8 = 1430$).
- **Exact packings on arbitrary point sets.** We reduce $\nu(P)$ to maximum independent set in the conflict graph on the $\binom{n}{3}$ triangles and solve

$$\max \sum_T x_T \quad \text{s.t.} \quad x_S + x_T \leq 1 \quad \text{for every conflicting pair,} \quad x_T \in \{0, 1\},$$

with HiGHS [4] through `scipy.optimize.milp`. The program has $\binom{n}{3}$ binary variables and $O(n^4)$ constraints; we solve it exactly for $n \leq 12$.

- **Maximal planar graph scan.** Enumerating all labelled maximal planar graphs on n vertices means testing $\binom{n}{3n-6}$ candidate graphs for planarity; $n = 7$ is about 10^5 candidates and takes a few minutes.

9 Limitations and open problems

1. **Conjecture 7.1 is unproved.** We verified it on 423 configurations for $n \leq 11$ (eleven structured families, exact integer programming) and on the families of Figure 1 for $n \leq 12$. A finite search is not a proof, and the second assertion, that equality forces convex position, is supported only computationally.
2. **The converse in Theorem 6.3** ($\nu = n - 2$ forces an Eulerian maximal planar graph) is verified for $n \leq 7$ and is not proved in general; see Remark 6.4.
3. **The convex value is a counting bound.** The upper bound $\lfloor (2n - 3)/3 \rfloor$ is a two-line count, so the content of Part 1 is the construction and the identification $f(n) = a(n - 2)$, not the value itself.

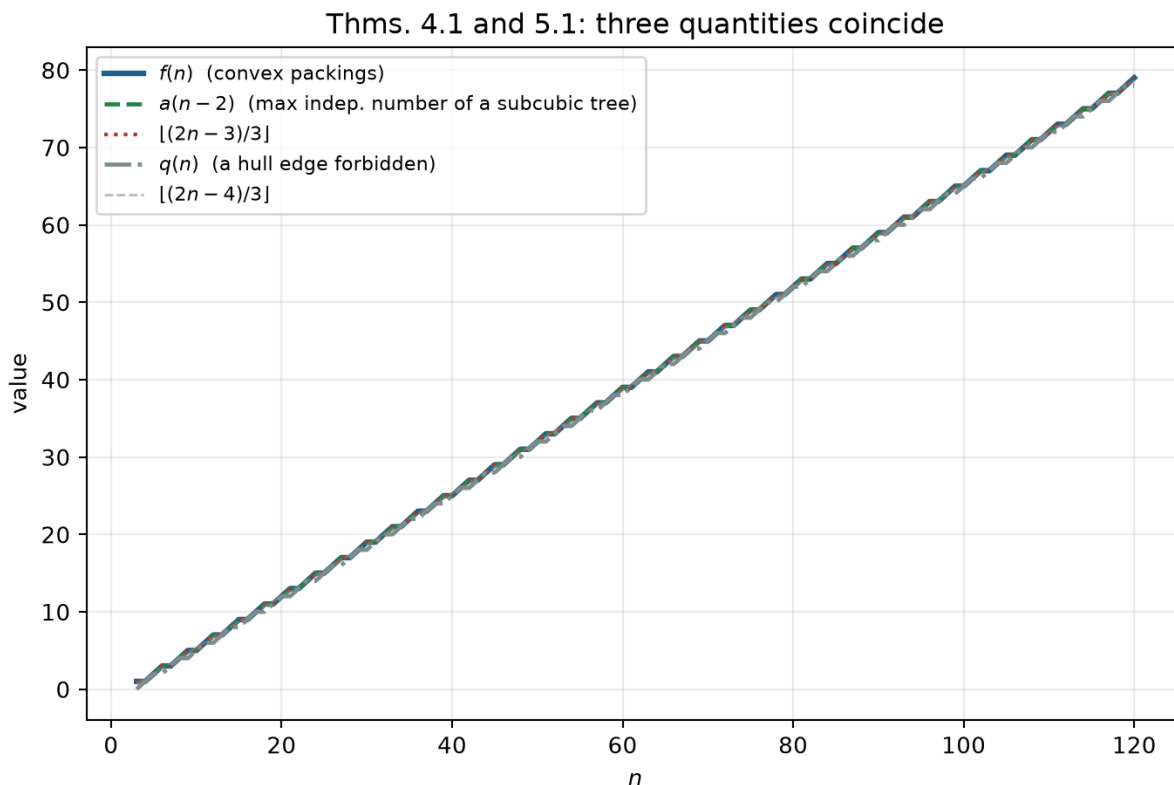


Figure 3: The three quantities of Theorem 5.1 coincide: the convex extremal function $f(n)$, the maximum independence number $a(n-2)$ of a subcubic tree, and the closed form $\lfloor (2n-3)/3 \rfloor$. Also shown is $q(n)$, the optimum when one hull edge is forbidden.

4. **No variants.** We did not investigate convex k -gons in place of triangles, weighted versions, or the effect of allowing triangle interiors to overlap.

10 Conclusion

We determined the largest number of pairwise non-crossing, edge-disjoint triangles spanned by n points in convex position: $\lfloor (2n-3)/3 \rfloor$. More interestingly, this geometric extremal function coincides, term by term, with the maximum independence number of a subcubic tree on $n-2$ vertices, itself equal to $\lfloor (2N+1)/3 \rfloor$; the two ingredients are that every 3-cycle of a convex-polygon triangulation bounds a face, and that every subcubic tree occurs as such a weak dual. For arbitrary point sets we proved the sharp bound $\nu(P) \leq n-2$ and determined which n admit extremal configurations, via Eulerian triangulations and a complete existence classification. What remains open here is whether convex position is the worst configuration; we state it precisely as Conjecture 7.1 and report our unsuccessful attempts to prove it.

References

- [1] O. Aichholzer, T. Hackl and M. Korman, “Packing plane spanning trees and paths in complete geometric graphs,” *Information Processing Letters* **124** (2017), 35–41.
- [2] F. Bernhart and P. Kainen, “The book thickness of a graph,” *Journal of Combinatorial Theory, Series B* **27** (1979), 320–331.

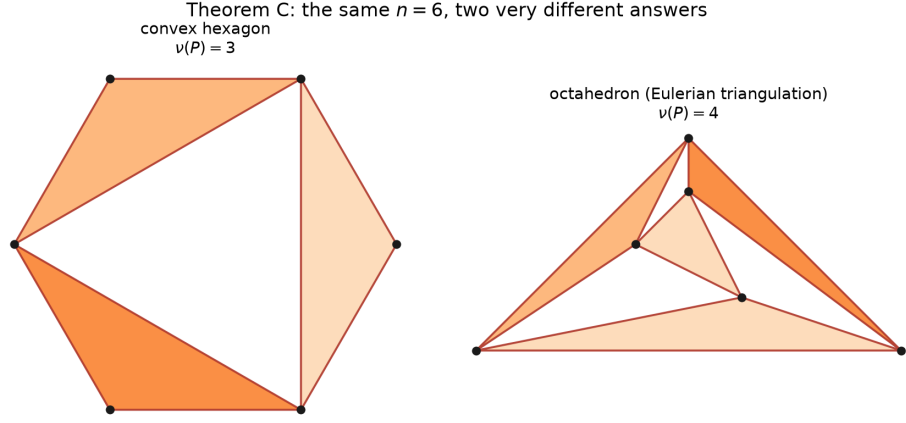


Figure 4: The same $n = 6$, two answers. Left: a convex hexagon supports $\nu = 3 = \lfloor (2n - 3)/3 \rfloor$. Right: a straight-line drawing of the octahedron supports $\nu = 4 = n - 2$.

- [3] Z. Füredi, T. Jiang and A. V. Kostochka, “Extremal problems for convex geometric hypergraphs and ordered hypergraphs,” *Canadian Journal of Mathematics* **73** (2021), 1648–1666.
- [4] Q. Huangfu and J. A. J. Hall, “Parallelizing the dual revised simplex method,” *Mathematical Programming Computation* **10** (2018), 119–142.
- [5] J. O’Neill and S. Spiro, “Saturation problems in convex geometric hypergraphs,” arXiv:2109.09931 (2021).
- [6] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, 2nd ed., Cambridge University Press, 2011.

A Reproduction

All experiments are deterministic. From a clean clone:

```
pip install -e .
python -m pytest tests -q
python experiments/run_all.py
python experiments/make_figures.py
```

Individual experiments, in increasing cost:

```
python experiments/exp1_convex.py      # f(n): DP and full enumeration, n<=10
python experiments/exp3_weakdual.py    # structural lemmas, exhaustively
python experiments/exp2_falsify.py     # Conjecture 7.1: hunt for counterexamples
python experiments/exp4_general.py     # Theorem 6.5: all maximal planar graphs, n<=7
```

The scan of all maximal planar graphs for $n = 7$ takes a few minutes; set the environment variable `PLANETRI_MAX_ENUM_N=6` to shorten it.

B Where the code lives

- `src/planetri/geom.py`
exact orientation and crossing predicates on exact points
- `src/planetri/solver.py`
conflict graph and exact maximiser via 0/1 programming
- `src/planetri/convex.py`
recurrence (3), closed form, constructive packings
- `src/planetri/trees.py`
independence numbers of subcubic trees, weak duals
- `tests/test_planetri.py`
the test suite